JAERYEONG KIM

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EDUCATION

ETH Zurich — September 2025 - Present
M.S. in Robotics, Systems, and Control — *Zurich, Switzerland*

Korea Advanced Institute of Science and Technology — March 2020 - August 2025
B.S. in Computer Science & Electrical and Electronic Engineering — *Daejeon, South Korea*

- Member of KAIST Presidential Fellowship (KPF)
- Summa Cum Laude

ETH Zurich — February 2024 - February 2025
ETH Zurich Exchange Program — *Zurich, Switzerland*

- GPA of 5.76/6.0, 59 ECTS credits across coursework and research.

University of California, Berkeley — June 2022 - August 2022
Berkeley Summer Session — *Berkeley, USA*

- Attended UC Berkeley through a full scholarship from the KAIST Presidential Fellowship.

WORK EXPERIENCE

ETH Zurich — Robotic Systems Lab — March 2026 – August 2026
Semester Project — Tactile Sensing for RL-based Locomotion — *Zurich, Switzerland*

- Investigated **whole-body tactile sensing** for reinforcement learning-based dynamic locomotion on the **ANYmal-D** quadruped robot, designing a dodging benchmark to compare tactile observation representations under one identical training recipe.
- Proposed **TaxelKV**, a taxel-grid representation matched to an FBG-based tactile skin, cutting impulse per contact from 1.30 N s to 0.45 N s and end-effector tracking error from 0.027 m to 0.011 m against the proprioceptive baseline.
- Deployed the policies on the real ANYmal-D carrying the tactile skin, showing that representations staying close to the real sensor's output are the most robust to sim-to-real gaps.

ETH Zurich — Computational Robotics Lab — September 2025 – Present
Spatio-temporal Motion Retargeting for Humanoids — *Zurich, Switzerland*

- Conduct research on **spatio-temporal motion retargeting (STMR)** and whole-body control for humanoid robots, focusing on kino-dynamically feasible motion generation from keypoint trajectories.
- Integrate sampling-based MPC and **Model Predictive Path Integral (MPPI)** methods using GPU-parallelized physics simulation (MuJoCo) to enable kino-dynamically feasible motion generation from keypoint trajectories.

ETH Zurich — Computational Robotics Lab — February 2024 - February 2025

Diversifying Gait Patterns using Unsupervised Skill Discovery — *Zurich, Switzerland*

- Implemented unsupervised skill discovery methods with discrete latent variables inspired by cutting-edge papers (DIAYN & DISDAIN) and continuous latent variables (ASE, METRA), to encode skills into a latent space for control.
- Trained a policy that allows for controlling height, gait patterns, pitch angle, and feet width in quadruped locomotion.
- Gained experience with IsaacLab, Python, ROS2, and Reinforcement Learning.

Learning and Demonstrating Multi-gait Quadruped Locomotion for Interactive Media Art Exhibition

- Developed a reinforcement learning (RL)-based locomotion controller capable of producing expressive gait patterns.
- Implemented and successfully deployed the RL-based controller for use in a live setting.
- Used for the AATB artistic studio's exhibition.

TEACHING

ETH Zurich 2026 – Present
Teaching Assistant *Zurich, Switzerland*

- Linear System Theory
- Planning and Decision Making for Autonomous Robots

STUDY PROJECTS

Real-to-Sim 6-DoF Object Pose Estimation (3D Vision) Spring 2026

- Estimated 6-DoF object pose from a single egocentric RGB video through a full manipulation sequence, generating simulation-ready trajectories for replay in Isaac Sim.
- Implemented the dynamics-consistent trajectory replay and a **residual RL** pipeline that jointly refines robot and object trajectories through simulated contact dynamics.
- Link: `https://hudela390.github.io/real2sim-6dpose/`

Visual Odometry Fall 2025

- Developed a monocular Visual Odometry pipeline for real-time camera pose estimation using KLT feature tracking, PnP-based localization, and landmark triangulation.
- Enhanced the baseline system with Local Bundle Adjustment, ground plane scale correction using SAM3 segmentation, and Sim(3)-based loop detection and closure, evaluated on the KITTI dataset.
- Link: `https://github.com/nicolejrkim/va25_project`

DCASE 2020 Task 2 Spring 2025

- Developed models for detecting anomalous machine sounds using unsupervised learning for condition monitoring.
- Implemented advanced architectures (e.g., NoisyArcMix, SpecNet) and applied feature normalization and model ensembling, improving AUC by 2.14%.

TECHNICAL STRENGTHS

Computer Languages	Python, C, C#, C++, PyTorch, ROS2	Fluent
	Scala, Rust, OCaml, F#	Intermediate
Simulation	NVIDIA Isaac Sim / Isaac Lab, MuJoCo	
Robot Hardware	Unitree Go2, Unitree G1, ANYmal	

HONORS

Scholarships

- Korea Presidential Science Scholarship (for undergraduate study). *2020 - 2024*
- KAIST Presidential Fellowship (KPF) (for undergraduate study). *2020 - 2024*
- Korea Foundation for Advanced Studies (KFAS) Scholarship (for undergraduate study). *2022 - 2023*
- Hanseong Nobel Youngsoojae Scholarship (for high school study). *2018 - 2019*

Awards

- Winner, Robot Competition — ETH Robotics Summer School (team 38° of Freedom) *2026*
- College of Engineering's Dean's List *Spring 2023*
- Samsung HumanTech Paper Award (High School Division), Gold Prize.
  "Graph kernel algorithm for compound property prediction, based on chemical structure similarity" *2020*

RESEARCH INTERESTS

Robotics & Control	MPC, Sampling-based MPC, MPPI, Whole-Body Control, Motion Retargeting
Reinforcement Learning	Deep RL, Unsupervised Skill Discovery, Sim-to-Real Transfer
Computer Vision	SLAM, Visual Odometry, Vision–Language–Action Models

RELEVANT COURSES

Systems & Control

· Linear System Theory	6.0
· Dynamic Programming and Optimal Control	6.0
· Planning and Decision Making for Autonomous Robots	6.0
· Robot Dynamics	6.0

Machine Learning & Reinforcement Learning

· Probabilistic Artificial Intelligence	5.5
· Machine Perception	5.5

LANGUAGE

English Highly proficient (C1); TOEFL(iBT) score: 112/120
Korean Native